

Part 1: Media Query Exercises

Exercise 1: Responsive Navigation Bar

Requirements

- Create a navigation bar with:
 - Home
 - About
 - Services
 - Contact
- Desktop (>768px): display menu items horizontally.
- Mobile (≤768px): display menu items vertically.

Concepts

- Flexbox
- Media Queries

Exercise 2: Responsive Student Profile Card

Requirements

Create a student card containing:

- Profile Image
- Name
- Branch
- Email

Desktop:

Image | Details

Mobile:

Image

Details

Concepts

- Flexbox
- Media Queries

Exercise 3: Responsive Gallery

Requirements

Display 8 images.

Desktop:

4 columns

Tablet:

2 columns

Mobile:

1 column

Concepts

- CSS Grid
- Media Queries

Exercise 4: Student Management Dashboard

Requirements

Create:

- Header
- Sidebar
- Main Content

Desktop:

Sidebar | Content

Mobile:

Content only

Hide sidebar on smaller screens.

Concepts

- Layout Design
- Media Queries

Part 2: Transition Exercises

Exercise 5: Animated Button

Requirements

Create a button.

Before Hover:

Blue background

After Hover:

Green background

Scale 1.1

Use:

transition: all 0.5s;

Exercise 6: Expanding Card

Requirements

Card size:

200px × 200px

On Hover:

250px × 250px

Use smooth transition.

Exercise 7: Navigation Hover Effect

Requirements

Menu links should:

- Change color
- Increase font size
- Underline smoothly

using:

transition

Exercise 8: Image Zoom Effect

Requirements

Display an image.

On Hover:

transform: scale(1.2);

with smooth transition.

Part 3: Animation Exercises

Exercise 9: Loading Spinner

Requirements

Create a circular loader.

Use:

@keyframes spin

Rotate continuously.

Exercise 10: Moving Ball

Requirements

A circle should move:

Left → Right → Left

continuously.

Use:

animation-iteration-count: infinite;

Exercise 11: Bouncing Text

Requirements

Display:

WELCOME

Letters should move up and down repeatedly.

Exercise 12: Traffic Signal Animation

Requirements

Create:

- Red Light
- Yellow Light
- Green Light

Automatically switch every 2 seconds.

Concepts

- Keyframes
- Animation Delay

Part 4: Combined Exercises

Exercise 13: Responsive Animated Login Form

Requirements

- Username
- Password
- Login Button

Features:

- Mobile Responsive
- Button Hover Transition
- Form Fade-In Animation

Exercise 14: Responsive Product Card

Requirements

Display:

- Product Image
- Name
- Price
- Buy Button

Features:

- Responsive Layout
- Image Zoom on Hover
- Button Transition

Exercise 15: Student Management System Homepage

Requirements

Create:

Header

- College Logo
- Menu

Hero Section

- Welcome Message
- Register Button

Features Section

- Student Registration
- Attendance
- Results

Footer

Features:

- Responsive using Media Queries
- Button Transitions
- Fade-In Animation

Responsive Navigation Bar

Objective

Design a responsive navigation menu.

Requirements

1. Create a webpage containing:
 - Logo
 - Home
 - About
 - Services
 - Contact
2. Desktop View
 - Menu items should appear horizontally.
 - Background color should be dark blue.

3. Mobile View (768px and below)

- Menu items should appear vertically.
- Each menu item should occupy full width.

Constraints

- Use Flexbox.
- Use Media Queries.
- No Bootstrap.

Expected Output

Desktop:

LOGO Home About Services Contact

Mobile:

LOGO

Home

About

Services

Contact

Student Profile Card

Objective

Create a responsive student profile card.

Requirements

Display:

- Student Photo
- Name
- Roll Number
- Branch
- Email

Desktop Layout:

Photo | Details

Mobile Layout:

Photo

Details

Constraints

- Width: 500px
- Border Radius: 10px
- Shadow Effect

Expected Output

```
-----  
| Photo | Student |  
|       | Details  |  
-----
```

Responsive Image Gallery

Objective

Create a responsive image gallery.

Requirements

Display:

- 8 Images

Desktop:

- 4 columns

Tablet:

- 2 columns

Mobile:

- 1 column

Constraints

- Use CSS Grid
- Image size should adjust automatically.

Expected Output

Desktop

```
1 2 3 4  
5 6 7 8
```

Tablet

```
1 2  
3 4  
5 6  
7 8
```

Mobile

```
1  
2  
3  
4  
5  
6  
7  
8
```